

CONTACT

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BASE	Pakistan · Remote worldwide

PRODUCT DEVELOPMENT

- Firmware Development
- Hardware Prototyping
- PCB Design
- 3D Enclosure Design

FIRMWARE

- C
- C++
- Python
- RTOS
- UART
- SPI
- I2C
- Interrupts

PCB & CAD TOOLS

- KiCad
- Altium Designer
- EasyEDA Pro
- Fusion 360
- MATLAB
- Arduino IDE
- Keil uVision
- Espressif IDF

PLATFORMS

- STM32
- ESP32 / S3 / C3 / C6
- ATmega328P
- Arduino
- Raspberry Pi 4

CERTIFICATIONS

Mastering KiCad
Coursera · Jul 2026

PCB Design with Altium Designer: Hands-On
Coursera

EDUCATION

BSc Electrical Engineering
Namal University, Mianwali
2023 – 2027

Pre-Engineering
Cambridge International Science College, Talagang
2021 – 2022

Tahir Ali

ELECTRONIC PRODUCT DEVELOPMENT ENGINEER · TA EMBEDDED SOLUTIONS

• Available for freelance product development projects

8	400+	5.0	12
Layers, largest board	Components, one board	Fiverr rating	Projects documented

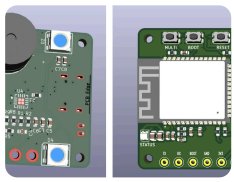
PROFILE

I take an electronic product from your idea to a complete, buildable design: I make it work first (firmware and a tested breadboard prototype), then design it professionally (KiCad schematic, PCB layout and a custom 3D-printable enclosure). You receive organised, editable engineering files that your chosen manufacturers can build from. I work as an individual freelance engineer, so you talk directly to the person doing the work.

HOW A PROJECT RUNS

01	02	03	04	05	06	07	08
Idea & requirements	Firmware	Breadboard prototype	Testing & debugging	KiCad schematic	PCB layout	3D enclosure	Engineering handover

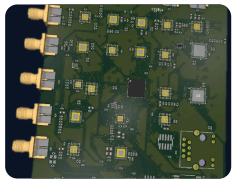
FEATURED CLIENT PROJECTS



Parcel-Tracking System: Beacon + Gateway

ESP32 · KICAD · FUSION 360

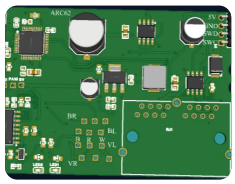
- Two ESP32 boards (ESP32-C3 Beacon, ESP32-S3 Gateway) with BLE + ESP-NOW, taken from the client's specification to manufacturing-ready files.
- My part: specification review and written design recommendations, KiCad schematics, two 4-layer PCBs, a 3D-printable Beacon enclosure and organised delivery packages.
- The client supplied the firmware and tested the prototypes.



Multi-Radio IoT Gateway, 8-Layer PCB

KICAD 10 · FIVERR

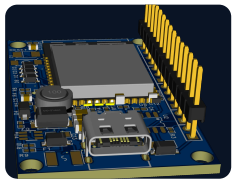
- Layout from the client's schematic and BOM: 400+ components on an STM32H7B3 board.
- Five RF sections (4× LR1121, 2× Si4463, AT86RF215), ten edge-mounted SMA connectors and an 8-output clock distribution network.



Load Cell Controller: Diagnostics & Repair

KICAD 10 · FIVERR

- No schematic provided: rebuilt it from the bare PCB and BOM (STM32F103, HX711, CAN daisy-chain).
- Diagnosed four field issues across a deployment of 30+ units and delivered a corrected KiCad revision and repair guide.



RejsaCAN v3.4 Redesign

EASYEDA PRO · FIVERR

- About 80 components in a 30 × 50 mm footprint: ESP32-S3, USB-C, SN65HVD230 CAN transceiver, microSD and a buck converter. Fabricated through JLCPCB.

"Tahir did an excellent job on my PCB project. He was very responsive, professional, and willing to make every requested adjustment until the final package was complete."

★★★★★ 5.0 · MAURIZIOMO, GERMANY · FIVERR REVIEW

LET'S WORK TOGETHER

Send me your idea, sketch, requirements or existing files. I reply within a day.

CALL +92 304 9037412

CHAT WhatsApp

HIRE Fiverr profile

WEB tahiriii-ai.github.io/ta-embedded-solutions

WHAT I DO NOT OFFER

I deliver design files and documentation. I do not manufacture or assemble PCBs, print enclosures, run mass production, ship products or buy components at my own expense.

EXPERIENCE

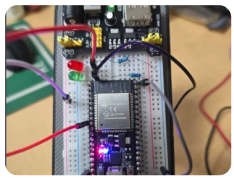
Freelance Electronic Product Development Engineer

TA EMBEDDED SOLUTIONS

Fiverr · Upwork · Freelancer.com · Remote, worldwide clients

- Take client ideas and specifications through component selection, schematic capture, PCB layout and enclosure CAD to organised engineering packages.
- Deliver manufacturer-ready files (Gerber, drill, BOM, CPL, STEP, STL) compatible with JLCPCB, PCBWay and OSHPark.
- Work across microcontroller boards, USB interface boards, RF boards and power electronics in KiCad, Altium Designer, EasyEDA Pro and Fusion 360.

HARDWARE PROTOTYPING



ESP32 breadboard prototype

I build and test the circuit on a breadboard with real components before any PCB is drawn: microcontroller, sensors, indicators and power, with firmware loaded and the behaviour checked on the bench.

WHAT I CAN DO FOR YOU

- Firmware (ESP32, STM32, Arduino)
- Schematic design
- Reverse engineering
- ERC / DRC verification
- Breadboard prototyping
- Multilayer PCB layout
- Design review and redesign
- Gerber / BOM preparation
- Hardware testing and debugging
- RF and power layout
- 3D enclosure design
- Documentation

HOW I WORK

01 **Understand** Idea, sketch, specification

02 **Firmware + prototype** Behaviour proven on a breadboard

03 **Test and validate** Debug before a PCB is drawn

04 **Schematic** KiCad schematic and BOM

05 **PCB layout** Placement, routing, DRC

06 **Enclosure** CAD around the real PCB

07 **Verification** ERC/DRC, design review

08 **Handover** Editable files and documentation

MORE PROJECTS

- **Custom ESP32 Dev Board.** 2-layer, USB-C, USBLC6 ESD, AP2112K 3.3 V LDO, RF keep-out.
- **STM32 GPS Tracker.** GPS antenna keep-out, microSD logging, dedicated ground plane.
- **Beacon enclosure.** Five printable parts designed around the finished PCB and battery.
- **USB to UART Converter.** CP2102, USB-C, ESD-protected data lines, TX/RX LEDs.
- **DC-DC Buck Converter.** Zero ERC/DRC errors, sized copper pour, 3D visualization.
- **Water Level Indicator.** BC547 stages, LEDs and buzzer, verified in KiCad and Altium.

EMBEDDED, AWARDS AND RESEARCH

- **Smart Blind Assistance System.** Winner. ESP32-C6 obstacle detection and alerts · Dec 2025
- **Edge AI Object Detection.** YOLO on Raspberry Pi with live camera · Mar 2025
- **STM32 UART Communication.** Interrupt-based PC control of peripherals · Jan 2026
- **Research poster, LUMS.** Quantization-aware compilation on RISC-V for edge AI · Mar 2026

Have an electronic product idea?

Send the idea, requirements or existing files and I will review them and suggest the next step.

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